



Shubhendra Pratap Singh

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PROFESSIONAL SUMMARY

Mechanical Engineer with 9+ years of experience in R&D, product design, rapid prototyping, and mechanical systems development for innovative applications. Expertise in developing new product concepts, verifying specifications, designing fixtures and testing processes, and specifying materials to ensure prototypes meet requirements. Proficient in SolidWorks, MATLAB, Python, and MS Office for technical documentation and process automation. Delivered cost-saving automation tools and contributed to bio-inspired designs, with publications in SAE/ASME.

PROFESSIONAL EXPERIENCE

Project Engineer

FEV India Pvt. Ltd. (Automotive Research Company), Pune, India

Nov 2017 – Apr 2021

- Developed macro-based automation tool merging 3 manual emissions test processes: reduced runtime from 30 min to 2-3 min, saving ~\$600k annually (tool still in production, 2021).
- Led piloted and chassis-dyno testing/calibration for Euro-6 diesel, SCR, and hybrid engines (Cummins, John Deere, Mahindra).
- Tools: ETAS INCA, Cummins Calterm, MATLAB/Simulink, AVL FTIR, Horiba MEXA.

Graduate Research Assistant – Digital Twins & Vehicle Design

Clemson University (Army GVSC Collaboration), Clemson, SC

Aug 2025 – Present

- Collaborated with U.S. Army GVSC on digital twin integration for ground vehicle systems.
- Evaluated 5 non-functional requirements (-ilities) for tracked military vehicles; delivered formal definitions and mathematical models (Maneuverability, Trafficability, etc.).
- Co-authored SAE WCX 2026 paper on effects of targeted instruction in vehicle design education.

Research Associate – Robotics & Automation

Club First Robotics (Robotic Company), Jaipur, India

Jul 2016 – Nov 2017

- Designed and validated Instant Water Purification & Vending System (IWPVS) for rural deployment.
- Built service robot mechanical structure, validated kinematics and dynamics via FEA.

Engineering Intern (Automotive Engineering)

Honda Cars India Ltd. (Car Manufacturer), Greater Noida, India

Jan 2016 – Jun 2016

- Designed "Master Jig" eliminating petrol/diesel changeover (38.5s loss to 0s); reduced operator count and advanced to supplier prototype.

EDUCATION

M.S., Mechanical Engineering (Thesis)

Clemson University, Clemson, SC

Aug 2023 – Dec 2025 (Expected)

GPA: 3.85 | Thesis: AI-Augmented Framework for Bio-Inspired Design (FFP Matching + LLM Replication)

- Developed an AI-augmented FFP framework that identifies and classifies analogy types in bio-inspired design-by-analogy (DbA) — enabling systematic biology-to-engineering translation.
- Long-term goal: scale framework to electromechanical, marine, and cybersecurity systems — enabling rapid, biology-to-engineering translation.

B.Tech, Mechanical Engineering

JECRC University, Jaipur, India

Aug 2012 – Aug 2016

Founder, JU Automotive Club | Led teams to national wins (Eco-Kart, Electric Baja) | Best Design Award

PUBLICATIONS & PATENTS

SAE WCX 2026 (Accepted)

- Effects of Targeted Instruction on Non-Functional Requirement Understanding
- Enhancing Track Vehicle Design Engineering Through Digital Twins

ASME IDETC-CIE 2025 (Accepted Poster)

- Identifying Adopted Strategies in Bio-Inspired Design via Critical Chain and FFP Matching

Patent Pending

- Self-Powered Whale-Bot Tagging Device (Non-invasive, kinetic energy harvesting)

SELECTED PROJECTS

- *Whale-Bot Tagging Device (2024–2025)*: Bio-inspired marine tracking tag; harvests whale motion for long-duration power; patent filing in process (supervised by Dr. Cameron Turner).
- *ECU IP-Theft Risk Study (2023–2024, Georgia Tech Collab)*: Demonstrated ECU benchmarking vulnerabilities; presented to cybersecurity faculty (Dr. Saman Zonouz and team).
- *Li-ion Battery On-Board Diagnostics (2023–2024)*: EKF-based state estimation with hybrid thermal-electrical modeling in MATLAB/Simulink.

TECHNICAL SKILLS

CAD & Prototyping: SolidWorks (Proficient), Fusion 360, Inventor, ANSYS | 3D Printing (FDM/SLA)

Programming: MS Word, Excel, PowerPoint | Programming: Python, MATLAB, Simulink, C++, R

Testing & Diagnostics: Engine Dynos, Climate Chambers, Emissions Analyzers, DFMEA, Functional Modeling, Tolerance Analysis, Critical Chain Modeling

Tools & Systems: ETAS INCA, Cummins Calterm, Horiba MEXA, SCADA, PLC

Frameworks: DFMEA, Functional Modeling, Tolerance Analysis, Critical Chain Modeling

PROFESSIONAL AFFILIATIONS

Member, SAE International

Research: CEDAR Lab, Deep Orange, VIPR-GS (Army Autonomy Prototyping)